

ABSTRACT REASONING (NON-VERBAL REASONING)

Full Question Bank
Questions, Solutions, and Explanations

SECTION: 01 Basic

--- README.md ---

100 Abstract Reasoning Practice Questions (Basic Level) with Solutions

This file contains 100 basic-level practice questions for Abstract Reasoning, designed to develop your ability to identify patterns, sequences, and relationships in visual and logical problems for fresher job aptitude tests, especially for AI/ML roles. Each question includes a description of the problem (e.g., number sequences, shape patterns, analogies) and a detailed solution explaining the reasoning process. Topics cover simple arithmetic sequences, geometric progressions, shape transformations, and basic analogies.

Number Sequence Questions (1?20)

1. Question: Find the next number in the sequence: 2, 4, 6, 8, __

Solution:

Each number increases by 2 (arithmetic sequence): $2 + 2 = 4$, $4 + 2 = 6$, $6 + 2 = 8$. Next: $8 + 2 = 10$.

Answer: 10

2. Question: Find the next number: 1, 3, 9, 27, __

Solution:

Each number is multiplied by 3 (geometric sequence): $1 \times 3 = 3$, $3 \times 3 = 9$, $9 \times 3 = 27$. Next: $27 \times 3 = 81$.

Answer: 81

3. Question: Find the next number: 5, 10, 15, 20, __

Solution:

Each number increases by 5: $5 + 5 = 10$, $10 + 5 = 15$, $15 + 5 = 20$. Next: $20 + 5 = 25$.

Answer: 25

4. Question: Find the next number: 1, 4, 16, 64, __

Solution:

Each number is multiplied by 4: $1 \times 4 = 4$, $4 \times 4 = 16$, $16 \times 4 = 64$. Next: $64 \times 4 = 256$.

Answer: 256

5. Question: Find the next number: 3, 6, 12, 24, __

Solution:

Each number is multiplied by 2: $3 \times 2 = 6$, $6 \times 2 = 12$, $12 \times 2 = 24$. Next: $24 \times 2 = 48$.

Answer: 48

6. Question: Find the next number: 10, 9, 8, 7, __

Solution:

Each number decreases by 1: $10 - 1 = 9$, $9 - 1 = 8$, $8 - 1 = 7$. Next: $7 - 1 = 6$.

Answer: 6

7. Question: Find the next number: 2, 5, 8, 11, __

Solution:

Each number increases by 3: $2 + 3 = 5$, $5 + 3 = 8$, $8 + 3 = 11$. Next: $11 + 3 = 14$.

Answer: 14

8. Question: Find the next number: 1, 2, 4, 8, __

Solution:

Each number is multiplied by 2: $1 \times 2 = 2$, $2 \times 2 = 4$, $4 \times 2 = 8$. Next: $8 \times 2 = 16$.

Answer: 16

9. Question: Find the next number: 100, 90, 80, 70, __

Solution:

Each number decreases by 10: $100 - 10 = 90$, $90 - 10 = 80$, $80 - 10 = 70$. Next: $70 - 10 = 60$.

Answer: 60

10. Question: Find the next number: 1, 5, 25, 125, __

Solution:

Each number is multiplied by 5: $1 \times 5 = 5$, $5 \times 5 = 25$, $25 \times 5 = 125$. Next: $125 \times 5 = 625$.

Answer: 625

11. Question: Find the next number: 4, 8, 12, 16, __

Solution:

Each number increases by 4: $4 + 4 = 8$, $8 + 4 = 12$, $12 + 4 = 16$. Next: $16 + 4 = 20$.

Answer: 20

12. Question: Find the next number: 2, 6, 18, 54, __

Solution:

Each number is multiplied by 3: $2 \times 3 = 6$, $6 \times 3 = 18$, $18 \times 3 = 54$. Next: $54 \times 3 = 162$.

Answer: 162

13. Question: Find the next number: 7, 14, 21, 28, __

Solution:

Each number increases by 7: $7 + 7 = 14$, $14 + 7 = 21$, $21 + 7 = 28$. Next: $28 + 7 = 35$.

Answer: 35

14. Question: Find the next number: 3, 9, 27, 81, __

Solution:

Each number is multiplied by 3: $3 \times 3 = 9$, $9 \times 3 = 27$, $27 \times 3 = 81$. Next: $81 \times 3 = 243$.

Answer: 243

15. Question: Find the next number: 50, 45, 40, 35, __

Solution:

Each number decreases by 5: $50 - 5 = 45$, $45 - 5 = 40$, $40 - 5 = 35$. Next: $35 - 5 = 30$.

Answer: 30

16. Question: Find the next number: 1, 3, 6, 10, __

Solution:

Differences: $3 - 1 = 2$, $6 - 3 = 3$, $10 - 6 = 4$. Differences increase by 1. Next difference = 5.

Next: $10 + 5 = 15$.

Answer: 15

17. Question: Find the next number: 2, 4, 8, 16, __

Solution:

Each number is multiplied by 2: $2 \times 2 = 4$, $4 \times 2 = 8$, $8 \times 2 = 16$. Next: $16 \times 2 = 32$.

Answer: 32

18. Question: Find the next number: 6, 12, 18, 24, __

Solution:

Each number increases by 6: $6 + 6 = 12$, $12 + 6 = 18$, $18 + 6 = 24$. Next: $24 + 6 = 30$.

Answer: 30

19. Question: Find the next number: 1, 4, 9, 16, __

Solution:

Numbers are squares: $1^2 = 1$, $2^2 = 4$, $3^2 = 9$, $4^2 = 16$. Next: $5^2 = 25$.

Answer: 25

20. Question: Find the next number: 8, 7, 6, 5, __

Solution:

Each number decreases by 1: $8 - 1 = 7$, $7 - 1 = 6$, $6 - 1 = 5$. Next: $5 - 1 = 4$.

Answer: 4

Shape Pattern Questions (21?40)

21. Question: A sequence of squares increases the number of small squares inside: 1, 4, 9, __. What is the next number?

Solution:

Numbers are squares: $1 = 1^2$, $4 = 2^2$, $9 = 3^2$. Next: $4^2 = 16$ small squares.

Answer: 16

22. Question: A triangle rotates 90° clockwise each step. After three rotations, what is the position?

Solution:

$90^\circ \times 3 = 270^\circ$ clockwise. The triangle is rotated 270° from the original position.

Answer: 270° clockwise

23. Question: A sequence of circles adds one circle each step: 1, 2, 3, 4, __. What is the next number?

Solution:

Each step adds 1: $1 + 1 = 2$, $2 + 1 = 3$, $3 + 1 = 4$. Next: $4 + 1 = 5$.

Answer: 5 circles

24. Question: A square is divided into smaller squares: 1, 4, 16, __. What is the next number?

Solution:

Each number is multiplied by 4: $1 \times 4 = 4$, $4 \times 4 = 16$. Next: $16 \times 4 = 64$.

Answer: 64

25. Question: A sequence of shapes alternates: Circle, Square, Circle, Square, __. What is the next shape?

Solution:

Pattern alternates: Circle, Square. After Square, the next is Circle.

Answer: Circle

26. Question: A pentagon rotates 72° clockwise each step. After two rotations, what is the position?

Solution:

$72^\circ \times 2 = 144^\circ$ clockwise.

Answer: 144° clockwise

27. Question: A sequence of triangles adds one triangle: 1, 2, 4, __. What is the next number?

Solution:

Each number is multiplied by 2: $1 \times 2 = 2$, $2 \times 2 = 4$. Next: $4 \times 2 = 8$.

Answer: 8 triangles

28. Question: A sequence of squares increases side length by 1: 1×1 , 2×2 , 3×3 , __. What is the next square?

Solution:

Side lengths: 1, 2, 3. Next: 4. Square is 4×4 .

Answer: 4×4

29. Question: A sequence alternates colors: Red, Blue, Red, Blue, __. What is the next color?

Solution:

Pattern alternates: Red, Blue. After Blue, the next is Red.

Answer: Red

30. Question: A circle is divided into sectors: 2, 4, 8, __. What is the next number?

Solution:

Each number is multiplied by 2: $2 \times 2 = 4$, $4 \times 2 = 8$. Next: $8 \times 2 = 16$.

Answer: 16 sectors

31. Question: A sequence of shapes: Star, Circle, Star, Circle, __. What is the next shape?

Solution:

Pattern alternates: Star, Circle. After Circle, the next is Star.

Answer: Star

32. Question: A square rotates 45° clockwise each step. After four rotations, what is the position?

Solution:

$45^\circ \times 4 = 180^\circ$ clockwise.

Answer: 180° clockwise

33. Question: A sequence of hexagons adds one side: 6, 7, 8, __. What is the next number?

Solution:

Each number increases by 1: $6 + 1 = 7$, $7 + 1 = 8$. Next: $8 + 1 = 9$.

Answer: 9 sides

34. Question: A sequence of shapes doubles: 1 triangle, 2 triangles, 4 triangles, __. What is the next number?

Solution:

Each number is multiplied by 2: $1 \times 2 = 2$, $2 \times 2 = 4$. Next: $4 \times 2 = 8$.

Answer: 8 triangles

35. Question: A sequence alternates: Filled Circle, Empty Circle, Filled Circle, __. What is the next shape?

Solution:

Pattern alternates: Filled, Empty. After Filled, the next is Empty Circle.

Answer: Empty Circle

36. Question: A rectangle increases length by 1: 2×1 , 3×1 , 4×1 , __. What is the next rectangle?

Solution:

Lengths: 2, 3, 4. Next: $4 + 1 = 5$. Rectangle is 5×1 .

Answer: 5×1

37. Question: A sequence of colors: Green, Yellow, Green, Yellow, __. What is the next color?

Solution:

Pattern alternates: Green, Yellow. After Yellow, the next is Green.

Answer: Green

38. Question: A triangle increases sides: 3, 4, 5, __. What is the next number?

Solution:

Each number increases by 1: $3 + 1 = 4$, $4 + 1 = 5$. Next: $5 + 1 = 6$.

Answer: 6 sides

39. Question: A sequence of shapes: Square, Triangle, Square, Triangle, __. What is the next

shape?

Solution:

Pattern alternates: Square, Triangle. After Triangle, the next is Square.

Answer: Square

40. Question: A circle increases radius by 1: $r=1$, $r=2$, $r=3$, _____. What is the next radius?

Solution:

Radii: 1, 2, 3. Next: $3 + 1 = 4$.

Answer: $r=4$

Analogy Questions (41?60)

41. Question: $2 : 4 :: 6 : \underline{\hspace{1cm}}$

Solution:

$2 \times 2 = 4$. Similarly, $6 \times 2 = 12$.

Answer: 12

42. Question: Circle : Square :: Triangle : ____

Solution:

Circle and Square are shapes. Triangle is a shape; the next shape with one more side is Quadrilateral.

Answer: Quadrilateral

43. Question: $1 : 3 :: 5 : \underline{\hspace{1cm}}$

Solution:

$1 \times 3 = 3$. Similarly, $5 \times 3 = 15$.

Answer: 15

44. Question: Red : Blue :: Yellow : ____

Solution:

Red and Blue are colors. Yellow is a color; the next color in a simple sequence (e.g., rainbow) is Green.

Answer: Green

45. Question: $3 : 9 :: 4 : \underline{\hspace{1cm}}$

Solution:

$3^2 = 9$. Similarly, $4^2 = 16$.

Answer: 16

46. Question: Star : Circle :: Pentagon : ____

Solution:

Star and Circle are shapes. Pentagon is a shape; the next shape with same sides (5) is Pentagon.

Answer: Pentagon

47. Question: $2 : 8 :: 3 : \underline{\hspace{1cm}}$

Solution:

$2^3 = 8$. Similarly, $3^3 = 27$.

Answer: 27

48. Question: Small : Large :: Short : ____

Solution:

Small and Large are size opposites. Short's opposite is Long.

Answer: Long

49. Question: 5 : 25 :: 6 : ____

Solution:

$5^2 = 25$. Similarly, $6^2 = 36$.

Answer: 36

50. Question: Triangle : 3 :: Square : ____

Solution:

Triangle has 3 sides. Square has 4 sides.

Answer: 4

51. Question: 1 : 2 :: 3 : ____

Solution:

$1 \times 2 = 2$. Similarly, $3 \times 2 = 6$.

Answer: 6

52. Question: Filled : Empty :: Black : ____

Solution:

Filled and Empty are opposites. Black's opposite is White.

Answer: White

53. Question: 4 : 16 :: 5 : ____

Solution:

$4^2 = 16$. Similarly, $5^2 = 25$.

Answer: 25

54. Question: Circle : 0 :: Triangle : ____

Solution:

Circle has 0 sides. Triangle has 3 sides.

Answer: 3

55. Question: 2 : 6 :: 4 : ____

Solution:

$2 \times 3 = 6$. Similarly, $4 \times 3 = 12$.

Answer: 12

56. Question: Red : Circle :: Blue : ____

Solution:

Red is a color, Circle is a shape. Blue is a color, so the shape remains Circle.

Answer: Circle

57. Question: $3 : 12 :: 5 : \underline{\hspace{1cm}}$

Solution:

$3 \times 4 = 12$. Similarly, $5 \times 4 = 20$.

Answer: 20

58. Question: Square : 4 :: Pentagon : $\underline{\hspace{1cm}}$

Solution:

Square has 4 sides. Pentagon has 5 sides.

Answer: 5

59. Question: $1 : 4 :: 2 : \underline{\hspace{1cm}}$

Solution:

$1 \times 4 = 4$. Similarly, $2 \times 4 = 8$.

Answer: 8

60. Question: Up : Down :: Left : $\underline{\hspace{1cm}}$

Solution:

Up and Down are opposite directions. Left's opposite is Right.

Answer: Right

Matrix Pattern Questions (61?80)

61. Question: Matrix: [1, 2, 3; 4, 5, 6; 7, 8, $\underline{\hspace{1cm}}$]. What is the missing number?

Solution:

Numbers increase by 1: 1, 2, 3, 4, 5, 6, 7, 8. Next: $8 + 1 = 9$.

Answer: 9

62. Question: Matrix: [2, 4, 6; 8, 10, 12; 14, 16, $\underline{\hspace{1cm}}$]. What is the missing number?

Solution:

Each row increases by 2. Third row: 14, 16, $16 + 2 = 18$.

Answer: 18

63. Question: Matrix: [1, 3, 5; 7, 9, 11; 13, 15, $\underline{\hspace{1cm}}$]. What is the missing number?

Solution:

Numbers increase by 2: 1, 3, 5, 7, 9, 11, 13, 15. Next: $15 + 2 = 17$.

Answer: 17

64. Question: Matrix: [1, 4, 16; 2, 8, 32; 3, 12, $\underline{\hspace{1cm}}$]. What is the missing number?

Solution:

Each row: $1 \times 4^?$, 1×4^1 , 1×4^2 ; $2 \times 4^?$, 2×4^1 , 2×4^2 ; $3 \times 4^?$, 3×4^1 , 3×4^2 .

Next: $3 \times 4^2 = 48$.

Answer: 48

65. Question: Matrix: [Circle, Square, Triangle; Square, Triangle, Circle; Triangle, Circle, ____]. What is the missing shape?

Solution:

Each column shifts shapes down: Circle ? Square ? Triangle; Square ? Triangle ? Circle; Triangle ? Circle ? Square.

Answer: Square

66. Question: Matrix: [2, 3, 4; 5, 6, 7; 8, 9, ____]. What is the missing number?

Solution:

Numbers increase by 1: 2, 3, 4, 5, 6, 7, 8, 9. Next: $9 + 1 = 10$.

Answer: 10

67. Question: Matrix: [1, 2, 4; 8, 16, 32; 64, 128, ____]. What is the missing number?

Solution:

Each row doubles: 1, 2, 4; 8, 16, 32; 64, 128, 256.

Answer: 256

68. Question: Matrix: [Red, Blue, Green; Blue, Green, Red; Green, Red, ____]. What is the missing color?

Solution:

Each column shifts colors down: Red ? Blue ? Green; Blue ? Green ? Red; Green ? Red ? Blue.

Answer: Blue

69. Question: Matrix: [3, 6, 9; 12, 15, 18; 21, 24, ____]. What is the missing number?

Solution:

Each row increases by 3. Third row: 21, 24, $24 + 3 = 27$.

Answer: 27

70. Question: Matrix: [1, 3, 9; 2, 6, 18; 4, 12, ____]. What is the missing number?

Solution:

Each row multiplies by 3: 1, 3, 9; 2, 6, 18; 4, 12, 36.

Answer: 36

71. Question: Matrix: [1, 4, 7; 10, 13, 16; 19, 22, ____]. What is the missing number?

Solution:

Numbers increase by 3: 1, 4, 7, 10, 13, 16, 19, 22. Next: $22 + 3 = 25$.

Answer: 25

72. Question: Matrix: [Star, Circle, Square; Circle, Square, Star; Square, Star, ____]. What is the missing shape?

Solution:

Each column shifts shapes down: Star ? Circle ? Square; Circle ? Square ? Star; Square ? Star ? Circle.

Answer: Circle

73. Question: Matrix: [2, 4, 8; 16, 32, 64; 128, 256, ____]. What is the missing number?

Solution:

Each row doubles: 2, 4, 8; 16, 32, 64; 128, 256, 512.

Answer: 512

74. Question: Matrix: [1, 5, 9; 13, 17, 21; 25, 29, ____]. What is the missing number?

Solution:

Numbers increase by 4: 1, 5, 9, 13, 17, 21, 25, 29. Next: $29 + 4 = 33$.

Answer: 33

75. Question: Matrix: [Filled, Empty, Filled; Empty, Filled, Empty; Filled, Empty, ____]. What is the missing state?

Solution:

Each column alternates: Filled ? Empty ? Filled; Empty ? Filled ? Empty; Filled ? Empty ? Filled.

Answer: Filled

76. Question: Matrix: [1, 2, 3; 4, 5, 6; 7, 8, ____]. What is the missing number?

Solution:

Numbers increase by 1: 1, 2, 3, 4, 5, 6, 7, 8. Next: $8 + 1 = 9$.

Answer: 9

77. Question: Matrix: [3, 9, 27; 2, 6, 18; 1, 3, ____]. What is the missing number?

Solution:

Each row multiplies by 3: 3, 9, 27; 2, 6, 18; 1, 3, 9.

Answer: 9

78. Question: Matrix: [Green, Yellow, Red; Yellow, Red, Green; Red, Green, ____]. What is the missing color?

Solution:

Each column shifts colors down: Green ? Yellow ? Red; Yellow ? Red ? Green; Red ? Green ? Yellow.

Answer: Yellow

79. Question: Matrix: [4, 8, 12; 16, 20, 24; 28, 32, ____]. What is the missing number?

Solution:

Each row increases by 4. Third row: 28, 32, $32 + 4 = 36$.

Answer: 36

80. Question: Matrix: [1, 4, 16; 2, 8, 32; 3, 12, ____]. What is the missing number?

Solution:

Each row: 1×4^0 , 1×4^1 , 1×4^2 ; 2×4^0 , 2×4^1 , 2×4^2 ; 3×4^0 , 3×4^1 , 3×4^2 .

Next: $3 \times 4^2 = 48$.

Answer: 48

Odd One Out Questions (81-100)

81. Question: Which is the odd one out: 2, 4, 6, 7, 8?

Solution:

2, 4, 6, 8 are even. 7 is odd.

Answer: 7

82. Question: Which is the odd one out: Circle, Square, Triangle, Pentagon, Star?

Solution:

Circle has no sides; others have sides.

Answer: Circle

83. Question: Which is the odd one out: 1, 3, 5, 7, 10?

Solution:

1, 3, 5, 7 are odd. 10 is even.

Answer: 10

84. Question: Which is the odd one out: Red, Blue, Green, Yellow, Square?

Solution:

Red, Blue, Green, Yellow are colors. Square is a shape.

Answer: Square

85. Question: Which is the odd one out: 4, 16, 25, 36, 49?

Solution:

16, 25, 36, 49 are perfect squares (4^2 , 5^2 , 6^2 , 7^2). 4 is 2^2 but smaller.

Answer: 4

86. Question: Which is the odd one out: Star, Circle, Square, Triangle, Hexagon?

Solution:

Star is not a regular polygon; others are.

Answer: Star

87. Question: Which is the odd one out: 2, 6, 12, 20, 30?

Solution:

Differences: 4, 6, 8, 10 increase by 2. 2 is not part of this pattern.

Answer: 2

88. Question: Which is the odd one out: Filled, Empty, Filled, Empty, Red?

Solution:

Filled, Empty are states. Red is a color.

Answer: Red

89. Question: Which is the odd one out: 1, 4, 9, 16, 24?

Solution:

1, 4, 9, 16 are squares (1^2 , 2^2 , 3^2 , 4^2). 24 is not.

Answer: 24

90. Question: Which is the odd one out: Triangle, Square, Pentagon, Circle, Hexagon?

Solution:

Circle has no sides; others have sides.

Answer: Circle

91. Question: Which is the odd one out: 3, 6, 9, 12, 14?

Solution:

3, 6, 9, 12 are multiples of 3. 14 is not.

Answer: 14

92. Question: Which is the odd one out: Blue, Green, Yellow, Red, Triangle?

Solution:

Blue, Green, Yellow, Red are colors. Triangle is a shape.

Answer: Triangle

93. Question: Which is the odd one out: 2, 8, 18, 32, 50?

Solution:

2, 8, 18, 32, 50 are 1^2+1 , 2^2+4 , 3^2+9 , 4^2+16 , 5^2+25 . 2 deviates slightly.

Answer: 2

94. Question: Which is the odd one out: Pentagon, Hexagon, Heptagon, Octagon, Circle?

Solution:

Circle has no sides; others are polygons.

Answer: Circle

95. Question: Which is the odd one out: 5, 10, 15, 20, 22?

Solution:

5, 10, 15, 20 are multiples of 5. 22 is not.

Answer: 22

96. Question: Which is the odd one out: Star, Filled Circle, Empty Circle, Square, Triangle?

Solution:

Star is not a regular shape; others are.

Answer: Star

97. Question: Which is the odd one out: 1, 2, 4, 7, 8?

Solution:

1, 2, 4, 8 are powers of 2 (2^0 , 2^1 , 2^2 , 2^3). 7 is not.

Answer: 7

98. Question: Which is the odd one out: Red, Blue, Green, Circle, Yellow?

Solution:

Red, Blue, Green, Yellow are colors. Circle is a shape.

Answer: Circle

99. Question: Which is the odd one out: 3, 9, 27, 81, 82?

Solution:

3, 9, 27, 81 are powers of 3 (3^1 , 3^2 , 3^3 , 3^4). 82 is not.

Answer: 82

100. Question: Which is the odd one out: Square, Triangle, Pentagon, Hexagon, Star?

Solution:

Star is not a regular polygon; others are.

Answer: Star

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SECTION: 02 Intermediate
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--- README.md ---

100 Abstract Reasoning Practice Questions (Intermediate Level) with Solutions

This file contains 100 intermediate-level practice questions for Abstract Reasoning, designed to strengthen your ability to identify complex patterns, sequences, and relationships in visual and logical problems for fresher job aptitude tests, especially for AI/ML roles. Each question includes a description of the problem (e.g., multi-step sequences, shape transformations, analogies, matrices) and a detailed solution explaining the reasoning process. Topics cover arithmetic and geometric sequences with multiple rules, shape patterns with combined attributes, abstract analogies, and matrix patterns with layered logic.

Number Sequence Questions (1?20)

1. Question: Find the next number: 1, 3, 6, 10, 15, __

Solution:

Differences: $3 - 1 = 2$, $6 - 3 = 3$, $10 - 6 = 4$, $15 - 10 = 5$. Differences increase by 1.

Next difference = 6. Next: $15 + 6 = 21$.

Answer: 21

2. Question: Find the next number: 2, 5, 11, 23, 47, __

Solution:

Each number is obtained by doubling the previous and adding 1: $2 \times 2 + 1 = 5$, $5 \times 2 + 1 = 11$,

$$11 \times 2 + 1 = 23, 23 \times 2 + 1 = 47.$$

$$\text{Next: } 47 \times 2 + 1 = 95.$$

Answer: 95

3. Question: Find the next number: 1, 4, 10, 22, 46, __

Solution:

Each number is obtained by doubling the previous and adding a sequence of 2, 2, 2, 2, ...:

$$1 \times 2 + 2 = 4, 4 \times 2 + 2 = 10, 10 \times 2 + 2 = 22, 22 \times 2 + 2 = 46.$$

$$\text{Next: } 46 \times 2 + 2 = 94.$$

Answer: 94

4. Question: Find the next number: 3, 6, 12, 21, 33, __

Solution:

Differences: 3, 6, 9, 12. Differences increase by 3. Next difference = 15.

$$\text{Next: } 33 + 15 = 48.$$

Answer: 48

5. Question: Find the next number: 2, 8, 18, 32, 50, __

Solution:

Numbers follow: $1^2 + 1, 2^2 + 4, 3^2 + 9, 4^2 + 16, 5^2 + 25$. Next: $6^2 + 36 = 36 + 36 = 72$.

Answer: 72

6. Question: Find the next number: 100, 96, 88, 76, 60, __

Solution:

Differences: -4, -8, -12, -16. Differences decrease by 4. Next difference = -20.

$$\text{Next: } 60 - 20 = 40.$$

Answer: 40

7. Question: Find the next number: 1, 2, 6, 24, 120, __

Solution:

Each number is multiplied by increasing integers: $1 \times 1 = 1, 1 \times 2 = 2, 2 \times 3 = 6, 6 \times 4 = 24, 24 \times 5 = 120$.

$$\text{Next: } 120 \times 6 = 720.$$

Answer: 720

8. Question: Find the next number: 5, 7, 11, 19, 35, __

Solution:

Each number is obtained by adding increasing primes: $5 + 2 = 7, 7 + 4 = 11, 11 + 8 = 19, 19 + 16 = 35$.

$$\text{Next prime} = 5 \text{ (doubled: } 32\text{)}. \text{ Next: } 35 + 32 = 67.$$

Answer: 67

9. Question: Find the next number: 4, 9, 25, 64, 144, __

Solution:

Numbers are: $2^2, 3^2, 5^2, 8^2, 12^2$. Bases: 2, 3, 5, 8, 12. Differences: 1, 2, 3, 4. Next base = $12 + 5 =$

17.

Next: $17^2 = 289$.

Answer: 289

10. Question: Find the next number: 1, 5, 14, 30, 54, __

Solution:

Differences: 4, 9, 16, 24. Second differences: 5, 7, 8. Next second difference ? 8.

Next difference = $24 + 8 = 32$. Next: $54 + 32 = 86$.

Answer: 86

11. Question: Find the next number: 3, 12, 27, 48, 75, __

Solution:

Numbers are: $1 \times 3, 2 \times 6, 3 \times 9, 4 \times 12, 5 \times 15$. Multipliers: 3, 6, 9, 12, 15 (+3).

Next: $6 \times 18 = 108$.

Answer: 108

12. Question: Find the next number: 2, 3, 7, 22, 78, __

Solution:

Each number is obtained by multiplying by increasing integers and adding 1:

$2 \times 1 + 1 = 3, 3 \times 2 + 1 = 7, 7 \times 3 + 1 = 22, 22 \times 4 + 1 = 89$.

Next: $78 \times 5 + 1 = 391$.

Answer: 391

13. Question: Find the next number: 10, 20, 35, 55, 80, __

Solution:

Differences: 10, 15, 20, 25. Differences increase by 5. Next difference = 30.

Next: $80 + 30 = 110$.

Answer: 110

14. Question: Find the next number: 1, 6, 16, 31, 51, __

Solution:

Differences: 5, 10, 15, 20. Differences increase by 5. Next difference = 25.

Next: $51 + 25 = 76$.

Answer: 76

15. Question: Find the next number: 4, 16, 36, 64, 100, __

Solution:

Numbers are: $2^2, 4^2, 6^2, 8^2, 10^2$. Next: $12^2 = 144$.

Answer: 144

16. Question: Find the next number: 2, 4, 10, 28, 82, __

Solution:

Each number is obtained by multiplying by 3 and subtracting 2:

$2 \times 3 - 2 = 4, 4 \times 3 - 2 = 10, 10 \times 3 - 2 = 28, 28 \times 3 - 2 = 82$.

Next: $82 \times 3 - 2 = 244$.

Answer: 244

17. Question: Find the next number: 1, 3, 7, 15, 31, __

Solution:

Each number is doubled and increased by 1: $1 \times 2 + 1 = 3$, $3 \times 2 + 1 = 7$, $7 \times 2 + 1 = 15$, $15 \times 2 + 1 = 31$.

Next: $31 \times 2 + 1 = 63$.

Answer: 63

18. Question: Find the next number: 5, 10, 25, 80, 305, __

Solution:

Each number is multiplied by increasing factors: $5 \times 2 = 10$, $10 \times 2.5 = 25$, $25 \times 3.2 = 80$, $80 \times 3.8125 = 305$.

Next factor ? 4.5. Next: $305 \times 4.5 = 1372.5$? 1373.

Answer: 1373

19. Question: Find the next number: 2, 5, 9, 14, 20, __

Solution:

Differences: 3, 4, 5, 6. Differences increase by 1. Next difference = 7.

Next: $20 + 7 = 27$.

Answer: 27

20. Question: Find the next number: 100, 98, 93, 85, 74, __

Solution:

Differences: -2, -5, -8, -11. Differences decrease by 3. Next difference = -14.

Next: $74 - 14 = 60$.

Answer: 60

Shape Pattern Questions (21?40)

21. Question: A square rotates 90° clockwise and adds one small square inside each step: 1, 2, 3, __. What is the next number of small squares?

Solution:

Small squares increase by 1: 1, 2, 3. Next: $3 + 1 = 4$.

Answer: 4 small squares

22. Question: A triangle alternates colors (Red, Blue) and rotates 120° clockwise. After three steps, what is the color and rotation?

Solution:

Colors: Red, Blue, Red. Rotation: $120^\circ \times 3 = 360^\circ = 0^\circ$.

Answer: Red, 0° rotation

23. Question: A sequence of circles doubles and alternates fill: Filled (1), Empty (2), Filled (4), __. What is the next state and number?

Solution:

Number doubles: 1, 2, 4, 8. Fill alternates: Filled, Empty, Filled, Empty.

Answer: Empty, 8 circles

24. Question: A pentagon increases sides by 1 and rotates 72° anticlockwise each step: Pentagon, Hexagon, Heptagon, __. What is the next shape and rotation?

Solution:

Sides: 5, 6, 7, 8 (Octagon). Rotation: $72^\circ \times 4 = 288^\circ$ anticlockwise.

Answer: Octagon, 288° anticlockwise

25. Question: A sequence alternates shapes and colors: Circle (Red), Square (Blue), Circle (Red), Square (Blue), __. What is the next shape and color?

Solution:

Pattern alternates: Circle (Red), Square (Blue). Next: Circle (Red).

Answer: Circle, Red

26. Question: A rectangle doubles length and halves width: 2×4 , 4×2 , 8×1 , __. What is the next rectangle?

Solution:

Length: 2, 4, 8, 16. Width: 4, 2, 1, 0.5. Next: 16×0.5 .

Answer: 16×0.5

27. Question: A sequence of triangles adds one and alternates fill: 1 (Filled), 2 (Empty), 3 (Filled), __. What is the next number and state?

Solution:

Number: 1, 2, 3, 4. Fill alternates: Filled, Empty, Filled, Empty.

Answer: 4 triangles, Empty

28. Question: A square increases side length by 2 and rotates 45° clockwise: 2×2 , 4×4 , 6×6 , __. What is the next square and rotation?

Solution:

Side: 2, 4, 6, 8. Rotation: 45° , 90° , 135° , 180° .

Answer: 8×8 , 180° clockwise

29. Question: A sequence cycles colors (Red, Blue, Green) and shapes (Circle, Square): Circle (Red), Square (Blue), Circle (Green), Square (Red), Circle (Blue), __. What is the next shape and color?

Solution:

Shapes alternate: Circle, Square. Colors cycle: Red, Blue, Green. Next: Square (Green).

Answer: Square, Green

30. Question: A circle is divided into sectors: 2, 4, 8, 16, __. What is the next number, and if it rotates 90° clockwise per step, what is the rotation?

Solution:

Sectors double: 2, 4, 8, 16, 32. Rotation: $90^\circ \times 5 = 450^\circ = 90^\circ$.

Answer: 32 sectors, 90° clockwise

31. Question: A sequence alternates shapes: Star (Filled), Circle (Empty), Star (Filled), Circle (Empty), __. What is the next shape and state?

Solution:

Pattern alternates: Star (Filled), Circle (Empty). Next: Star (Filled).

Answer: Star, Filled

32. Question: A hexagon rotates 60° anticlockwise and adds one side: Hexagon, Heptagon, Octagon, __. What is the next shape and rotation?

Solution:

Sides: 6, 7, 8, 9 (Nonagon). Rotation: $60^\circ \times 4 = 240^\circ$ anticlockwise.

Answer: Nonagon, 240° anticlockwise

33. Question: A sequence of squares adds 1 small square and alternates colors: 1 (Red), 2 (Blue), 3 (Red), __. What is the next number and color?

Solution:

Squares: 1, 2, 3, 4. Colors alternate: Red, Blue, Red, Blue.

Answer: 4 squares, Blue

34. Question: A triangle doubles sides and rotates 120° clockwise: 3, 6, 12, __. What is the next number of sides and rotation?

Solution:

Sides: 3, 6, 12, 24. Rotation: $120^\circ \times 4 = 480^\circ = 120^\circ$.

Answer: 24 sides, 120° clockwise

35. Question: A sequence cycles shapes (Circle, Square, Triangle) and fills: Circle (Filled), Square (Empty), Triangle (Filled), Circle (Empty), Square (Filled), __. What is the next shape and state?

Solution:

Shapes cycle: Circle, Square, Triangle. Fill cycles: Filled, Empty, Filled. Next: Triangle (Empty).

Answer: Triangle, Empty

36. Question: A rectangle increases width by 1 and doubles length: 2×1 , 4×2 , 8×3 , __. What is the next rectangle?

Solution:

Length: 2, 4, 8, 16. Width: 1, 2, 3, 4. Next: 16×4 .

Answer: 16×4

37. Question: A sequence of circles adds 2 and alternates fill: 2 (Filled), 4 (Empty), 6 (Filled), __. What is the next number and state?

Solution:

Number: 2, 4, 6, 8. Fill alternates: Filled, Empty, Filled, Empty.

Answer: 8 circles, Empty

38. Question: A pentagon increases sides by 2 and rotates 36° clockwise: 5, 7, 9, __. What is the next shape and rotation?

Solution:

Sides: 5, 7, 9, 11 (11-gon). Rotation: $36^\circ \times 4 = 144^\circ$.

Answer: 11-gon, 144° clockwise

39. Question: A sequence alternates shapes and colors: Triangle (Red), Square (Blue), Triangle (Red), Square (Blue), ___. What is the next shape and color?

Solution:

Pattern alternates: Triangle (Red), Square (Blue). Next: Triangle (Red).

Answer: Triangle, Red

40. Question: A circle increases radius by 2 and rotates 45° anticlockwise: $r=2$, $r=4$, $r=6$, ___. What is the next radius and rotation?

Solution:

Radius: 2, 4, 6, 8. Rotation: $45^\circ \times 4 = 180^\circ$ anticlockwise.

Answer: $r=8$, 180° anticlockwise

Analogy Questions (41?60)

41. Question: $2 : 6 :: 5 : \underline{\hspace{1cm}}$

Solution:

$2 \times 3 = 6$. Similarly, $5 \times 3 = 15$.

Answer: 15

42. Question: Circle (Red) : Square (Blue) :: Triangle (Green) : ___

Solution:

Circle to Square changes shape, Red to Blue changes color. Triangle to Square, Green to Red (next color).

Answer: Square (Red)

43. Question: $3 : 12 :: 4 : \underline{\hspace{1cm}}$

Solution:

$3 \times 4 = 12$. Similarly, $4 \times 4 = 16$.

Answer: 16

44. Question: Filled Circle : Empty Square :: Filled Triangle : ___

Solution:

Filled to Empty changes state, Circle to Square changes shape. Filled Triangle to Empty Square.

Answer: Empty Square

45. Question: $5 : 30 :: 6 : \underline{\hspace{1cm}}$

Solution:

$5 \times (5 + 1) = 30$. Similarly, $6 \times (6 + 1) = 42$.

Answer: 42

46. Question: Star (3) : Circle (0) :: Pentagon (5) : ____

Solution:

Star (3 sides) to Circle (0 sides). Pentagon (5 sides) to Circle (0 sides).

Answer: Circle (0)

47. Question: 2 : 10 :: 3 : ____

Solution:

$2 \times (2 + 3) = 10$. Similarly, $3 \times (3 + 3) = 18$.

Answer: 18

48. Question: Red : Green :: Blue : ____

Solution:

Red to Green is a color shift. Blue to Yellow (next in sequence: Red, Green, Blue, Yellow).

Answer: Yellow

49. Question: 4 : 20 :: 5 : ____

Solution:

$4 \times 5 = 20$. Similarly, $5 \times 5 = 25$.

Answer: 25

50. Question: Triangle (3) : Square (4) :: Pentagon (5) : ____

Solution:

Triangle (3 sides) to Square (4 sides). Pentagon (5 sides) to Hexagon (6 sides).

Answer: Hexagon (6)

51. Question: 1 : 4 :: 3 : ____

Solution:

$1 \times 2 + 2 = 4$. Similarly, $3 \times 2 + 2 = 8$.

Answer: 8

52. Question: Filled : Empty :: Up : ____

Solution:

Filled to Empty is opposite. Up to Down is opposite.

Answer: Down

53. Question: 6 : 42 :: 7 : ____

Solution:

$6 \times (6 + 1) = 42$. Similarly, $7 \times (7 + 1) = 56$.

Answer: 56

54. Question: Circle (0) : Triangle (3) :: Square (4) : ____

Solution:

Circle (0 sides) to Triangle (3 sides). Square (4 sides) to Heptagon (7 sides, +3).

Answer: Heptagon (7)

55. Question: 2 : 12 :: 4 : ____

Solution:

$2 \times (2 + 4) = 12$. Similarly, $4 \times (4 + 4) = 32$.

Answer: 32

56. Question: Red Circle : Blue Square :: Green Triangle : ____

Solution:

Red to Blue, Circle to Square. Green to Red, Triangle to Square.

Answer: Red Square

57. Question: 3 : 15 :: 5 : ____

Solution:

$3 \times 5 = 15$. Similarly, $5 \times 5 = 25$.

Answer: 25

58. Question: Square (Filled) : Circle (Empty) :: Triangle (Filled) : ____

Solution:

Square to Circle changes shape, Filled to Empty changes state. Triangle to Circle, Filled to Empty.

Answer: Circle (Empty)

59. Question: 4 : 24 :: 6 : ____

Solution:

$4 \times (4 + 2) = 24$. Similarly, $6 \times (6 + 2) = 48$.

Answer: 48

60. Question: Pentagon (5) : Hexagon (6) :: Heptagon (7) : ____

Solution:

Pentagon (5 sides) to Hexagon (6 sides). Heptagon (7 sides) to Octagon (8 sides).

Answer: Octagon (8)

Matrix Pattern Questions (61-80)

61. Question: Matrix: [1, 4, 10; 2, 7, 16; 3, 10, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 4, 7, 10 (+3); 10, 16, x (+6). Next: $16 + 6 = 22$.

Answer: 22

62. Question: Matrix: [2, 6, 12; 3, 12, 36; 4, 24, ____]. What is the missing number?

Solution:

Each row: 2×1 , 2×3 , 2×6 ; 3×1 , 3×4 , 3×12 ; 4×1 , 4×6 , 4×18 .

Next: $4 \times 18 = 72$.

Answer: 72

63. Question: Matrix: [Red, Blue, Green; Blue, Green, Red; Green, Red, ____]. What is the missing

color?

Solution:

Each column shifts down: Red ? Blue ? Green; Blue ? Green ? Red; Green ? Red ? Blue.

Answer: Blue

64. Question: Matrix: [1, 3, 6; 8, 10, 12; 15, 17, ____]. What is the missing number?

Solution:

Numbers increase by differences: 1, 3 (+2), 6 (+3), 8 (+2), 10 (+2), 12 (+2), 15 (+3), 17 (+2), x (+2).

Next: $17 + 2 = 19$.

Answer: 19

65. Question: Matrix: [Circle, Square, Triangle; Square, Triangle, Circle; Triangle, Circle, ____]. What is the missing shape?

Solution:

Each column shifts down: Circle ? Square ? Triangle; Square ? Triangle ? Circle; Triangle ? Circle ? Square.

Answer: Square

66. Question: Matrix: [2, 8, 18; 3, 12, 24; 4, 16, ____]. What is the missing number?

Solution:

Each row: 2×1^2 , 2×2^2 , 2×3^2 ; 3×1^2 , 3×2^2 , 3×3^2 ; 4×1^2 , 4×2^2 , 4×3^2 .

Next: $4 \times 3^2 = 36$.

Answer: 36

67. Question: Matrix: [1, 5, 11; 2, 8, 17; 3, 11, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 5, 8, 11 (+3); 11, 17, x (+6). Next: $17 + 6 = 23$.

Answer: 23

68. Question: Matrix: [Filled, Empty, Filled; Empty, Filled, Empty; Filled, Empty, ____]. What is the missing state?

Solution:

Each column alternates: Filled ? Empty ? Filled; Empty ? Filled ? Empty; Filled ? Empty ? Filled.

Answer: Filled

69. Question: Matrix: [4, 12, 24; 6, 18, 36; 8, 24, ____]. What is the missing number?

Solution:

Each row: 4×1 , 4×3 , 4×6 ; 6×1 , 6×3 , 6×6 ; 8×1 , 8×3 , 8×6 .

Next: $8 \times 6 = 48$.

Answer: 48

70. Question: Matrix: [1, 4, 16; 2, 8, 32; 3, 12, ____]. What is the missing number?

Solution:

Each row: $1 \times 4?$, 1×4^1 , 1×4^2 ; $2 \times 4?$, 2×4^1 , 2×4^2 ; $3 \times 4?$, 3×4^1 , 3×4^2 .

Next: $3 \times 4^2 = 48$.

Answer: 48

71. Question: Matrix: [Star, Circle, Square; Circle, Square, Star; Square, Star, ____]. What is the missing shape?

Solution:

Each column shifts down: Star ? Circle ? Square; Circle ? Square ? Star; Square ? Star ? Circle.

Answer: Circle

72. Question: Matrix: [2, 10, 26; 3, 15, 39; 4, 20, ____]. What is the missing number?

Solution:

Each row: $2 \times (1 + 4)$, $2 \times (5 + 0)$, $2 \times (13 - 0)$; $3 \times (1 + 4)$, 3×5 , 3×13 ; $4 \times (1 + 4)$, 4×5 , $4 \times$
13.

Next: $4 \times 13 = 52$.

Answer: 52

73. Question: Matrix: [1, 6, 14; 2, 8, 17; 3, 10, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 6, 8, 10 (+2); 14, 17, x (+3). Next: $17 + 3 = 20$.

Answer: 20

74. Question: Matrix: [Red, Green, Blue; Green, Blue, Red; Blue, Red, ____]. What is the missing color?

Solution:

Each column shifts down: Red ? Green ? Blue; Green ? Blue ? Red; Blue ? Red ? Green.

Answer: Green

75. Question: Matrix: [3, 9, 18; 4, 12, 24; 5, 15, ____]. What is the missing number?

Solution:

Each row: 3×1 , 3×3 , 3×6 ; 4×1 , 4×3 , 4×6 ; 5×1 , 5×3 , 5×6 .

Next: $5 \times 6 = 30$.

Answer: 30

76. Question: Matrix: [1, 3, 7; 9, 11, 15; 17, 19, ____]. What is the missing number?

Solution:

Numbers increase by differences: 1, 3 (+2), 7 (+4), 9 (+2), 11 (+2), 15 (+4), 17 (+2), 19 (+2), x
(+4).

Next: $19 + 4 = 23$.

Answer: 23

77. Question: Matrix: [2, 8, 20; 3, 12, 30; 4, 16, ____]. What is the missing number?

Solution:

Each row: 2×1 , 2×4 , 2×10 ; 3×1 , 3×4 , 3×10 ; 4×1 , 4×4 , 4×10 .

Next: $4 \times 10 = 40$.

Answer: 40

78. Question: Matrix: [Filled Circle, Empty Square, Filled Triangle; Empty Square, Filled Triangle, Filled Circle; Filled Triangle, Filled Circle, ____]. What is the missing shape?

Solution:

Each column shifts down: Circle ? Square ? Triangle; Square ? Triangle ? Circle; Triangle ? Circle ? Square.

Answer: Empty Square

79. Question: Matrix: [4, 16, 36; 5, 20, 45; 6, 24, ____]. What is the missing number?

Solution:

Each row: 4×1^2 , 4×2^2 , 4×3^2 ; 5×1^2 , 5×2^2 , 5×3^2 ; 6×1^2 , 6×2^2 , 6×3^2 .

Next: $6 \times 3^2 = 54$.

Answer: 54

80. Question: Matrix: [1, 5, 13; 2, 8, 20; 3, 11, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 5, 8, 11 (+3); 13, 20, x (+7). Next: $20 + 7 = 27$.

Answer: 27

Odd One Out Questions (81?100)

81. Question: Which is the odd one out: 3, 9, 18, 27, 36?

Solution:

3, 9, 27, 36 are multiples of 3. 18 is not divisible by 3 evenly in this context.

Answer: 18

82. Question: Which is the odd one out: Circle (Red), Square (Blue), Triangle (Green), Pentagon (Red), Star (Blue)?

Solution:

Circle, Square, Triangle, Pentagon are polygons. Star is not.

Answer: Star

83. Question: Which is the odd one out: 2, 6, 12, 20, 30?

Solution:

Differences: 4, 6, 8, 10. 2 does not fit the pattern of increasing differences.

Answer: 2

84. Question: Which is the odd one out: Filled Circle, Empty Square, Filled Triangle, Empty Pentagon, Red Hexagon?

Solution:

Filled, Empty are states. Red is a color.

Answer: Red Hexagon

85. Question: Which is the odd one out: 4, 16, 36, 64, 81?

Solution:

4, 16, 36, 64 are squares of even numbers (2^2 , 4^2 , 6^2 , 8^2). 81 is 9^2 (odd).

Answer: 81

86. Question: Which is the odd one out: Star, Circle, Square, Triangle, Filled Pentagon?

Solution:

Filled Pentagon has a state; others are shapes only.

Answer: Filled Pentagon

87. Question: Which is the odd one out: 1, 5, 14, 30, 54?

Solution:

Differences: 4, 9, 16, 24. 54 does not fit the pattern of increasing differences.

Answer: 54

88. Question: Which is the odd one out: Red, Blue, Green, Yellow, Empty?

Solution:

Red, Blue, Green, Yellow are colors. Empty is a state.

Answer: Empty

89. Question: Which is the odd one out: 2, 8, 20, 40, 70?

Solution:

Differences: 6, 12, 20, 30. 70 does not fit the pattern.

Answer: 70

90. Question: Which is the odd one out: Triangle, Square, Pentagon, Circle, Hexagon?

Solution:

Circle has no sides; others are polygons.

Answer: Circle

91. Question: Which is the odd one out: 3, 12, 27, 48, 76?

Solution:

Numbers are: 1×3 , 2×6 , 3×9 , 4×12 , 5×15 . $76 \neq 5 \times 15$.

Answer: 76

92. Question: Which is the odd one out: Blue Circle, Red Square, Green Triangle, Yellow Pentagon, Star?

Solution:

Star has no color; others have colors.

Answer: Star

93. Question: Which is the odd one out: 5, 10, 25, 80, 305?

Solution:

Each number is multiplied by increasing factors. 305 does not fit exactly.

Answer: 305

94. Question: Which is the odd one out: Pentagon, Hexagon, Heptagon, Octagon, Filled Circle?

Solution:

Filled Circle has a state; others are shapes.

Answer: Filled Circle

95. Question: Which is the odd one out: 2, 5, 9, 14, 22?

Solution:

Differences: 3, 4, 5, 8. 22 does not fit the pattern.

Answer: 22

96. Question: Which is the odd one out: Filled Star, Empty Circle, Filled Square, Empty Triangle, Green Pentagon?

Solution:

Green is a color; others are states.

Answer: Green Pentagon

97. Question: Which is the odd one out: 1, 3, 7, 15, 30?

Solution:

Each number doubles and adds 1. 30 does not fit.

Answer: 30

98. Question: Which is the odd one out: Red, Blue, Circle, Green, Yellow?

Solution:

Circle is a shape; others are colors.

Answer: Circle

99. Question: Which is the odd one out: 4, 16, 36, 100, 144?

Solution:

4, 16, 36, 100 are squares of even numbers (2^2 , 4^2 , 6^2 , 10^2). 144 is 12^2 but fits.

Answer: 144 (contextual)

100. Question: Which is the odd one out: Square (Red), Triangle (Blue), Pentagon (Green), Hexagon (Yellow), Star (Filled)?

Solution:

Star has a state (Filled); others have colors.

Answer: Star

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SECTION: 03 Advance

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--- README.md ---

100 Abstract Reasoning Practice Questions (Advanced Level) with Solutions

This file contains 100 advanced-level practice questions for Abstract Reasoning, designed to rigorously test your ability to identify intricate patterns, sequences, and relationships in visual and logical problems for fresher job aptitude tests, especially for AI/ML roles. Each question includes a detailed description of the problem (e.g., multi-layered sequences, complex shape transformations, abstract analogies, composite matrices) and a comprehensive solution explaining the reasoning process. Topics cover multi-step arithmetic and geometric sequences, shape patterns with interdependent attributes, abstract analogies with nuanced relationships, matrix patterns with layered logic, and odd one out questions requiring deep analysis.

Number Sequence Questions (1?20)

1. Question: Find the next number: 1, 2, 6, 24, 120, 720, __

Solution:

Each number is multiplied by increasing integers (factorial): $1 \times 1 = 1$, $1 \times 2 = 2$, $2 \times 3 = 6$, $6 \times 4 = 24$, $24 \times 5 = 120$, $120 \times 6 = 720$.

Next: $720 \times 7 = 5040$.

Answer: 5040

2. Question: Find the next number: 3, 7, 18, 44, 94, 182, __

Solution:

Differences: 4, 11, 26, 50, 88. Second differences: 7, 15, 24, 38. Third differences: 8, 9, 14. Next third difference ? 22.

Second difference = $38 + 22 = 60$. First difference = $88 + 60 = 148$. Next: $182 + 148 = 330$.

Answer: 330

3. Question: Find the next number: 2, 5, 14, 41, 122, 365, __

Solution:

Each number is multiplied by 3 and subtracted by 1: $2 \times 3 - 1 = 5$, $5 \times 3 - 1 = 14$, $14 \times 3 - 1 = 41$, $41 \times 3 - 1 = 122$, $122 \times 3 - 1 = 365$.

Next: $365 \times 3 - 1 = 1094$.

Answer: 1094

4. Question: Find the next number: 1, 4, 13, 40, 121, 364, __

Solution:

Each number is multiplied by 3 and increased by 1: $1 \times 3 + 1 = 4$, $4 \times 3 + 1 = 13$, $13 \times 3 + 1 = 40$, $40 \times 3 + 1 = 121$, $121 \times 3 + 1 = 364$.

Next: $364 \times 3 + 1 = 1093$.

Answer: 1093

5. Question: Find the next number: 5, 10, 22, 49, 103, 208, __

Solution:

Each number is approximately doubled and increased by a sequence: $5 \times 2 = 10$, $10 \times 2 + 2 = 22$, $22 \times 2 + 5 = 49$, $49 \times 2 + 5 = 103$, $103 \times 2 + 2 = 208$.

Pattern alternates +2, +5. Next: $208 \times 2 + 5 = 421$.

Answer: 421

6. Question: Find the next number: 100, 95, 85, 70, 50, 25, __

Solution:

Differences: -5, -10, -15, -20, -25. Differences decrease by 5. Next difference = -30.

Next: $25 - 30 = -5$.

Answer: -5

7. Question: Find the next number: 2, 6, 15, 31, 56, 92, __

Solution:

Differences: 4, 9, 16, 25, 36 (squares: $2^2, 3^2, 4^2, 5^2, 6^2$). Next difference = $7^2 = 49$.

Next: $92 + 49 = 141$.

Answer: 141

8. Question: Find the next number: 1, 3, 10, 32, 94, 250, __

Solution:

Each number follows: $1 \times 3 - 0 = 3$, $3 \times 3 + 1 = 10$, $10 \times 3 + 2 = 32$, $32 \times 3 - 2 = 94$, $94 \times 3 - 32 = 250$.

Additions/subtractions: 0, +1, +2, -2, -32. Next: +62 (alternating signs, squares). Next: $250 \times 3 + 62 = 812$.

Answer: 812

9. Question: Find the next number: 4, 12, 30, 66, 132, 240, __

Solution:

Differences: 8, 18, 36, 66, 108. Second differences: 10, 18, 30, 42. Third differences: 8, 12, 12. Next: $42 + 12 = 54$.

First difference = $108 + 54 = 162$. Next: $240 + 162 = 402$.

Answer: 402

10. Question: Find the next number: 1, 5, 16, 44, 104, 224, __

Solution:

Differences: 4, 11, 28, 60, 120. Second differences: 7, 17, 32, 60. Third differences: 10, 15, 28. Next ? 48.

Second difference = $60 + 48 = 108$. First difference = $120 + 108 = 228$. Next: $224 + 228 = 452$.

Answer: 452

11. Question: Find the next number: 3, 9, 22, 46, 88, 158, __

Solution:

Differences: 6, 13, 24, 42, 70. Second differences: 7, 11, 18, 28. Third differences: 4, 7, 10. Next ? 16.

Second difference = $28 + 16 = 44$. First difference = $70 + 44 = 114$. Next: $158 + 114 = 272$.

Answer: 272

12. Question: Find the next number: 2, 4, 11, 29, 74, 182, __

Solution:

Each number follows: $2 \times 2 - 0 = 4$, $4 \times 3 - 1 = 11$, $11 \times 3 - 4 = 29$, $29 \times 3 - 13 = 74$, $74 \times 3 - 40 = 182$.

Subtractions: 0, 1, 4, 13, 40 (Fibonacci-like). Next: 104. Next: $182 \times 3 - 104 = 442$.

Answer: 442

13. Question: Find the next number: 10, 20, 35, 65, 140, 350, __

Solution:

Each number is multiplied by increasing factors: $10 \times 2 = 20$, $20 \times 1.75 = 35$, $35 \times 1.857 = 65$, $65 \times 2.154 = 140$, $140 \times 2.5 = 350$.

Next factor ? 2.8. Next: $350 \times 2.8 = 980$.

Answer: 980

14. Question: Find the next number: 1, 6, 18, 45, 103, 224, __

Solution:

Differences: 5, 12, 27, 58, 121. Second differences: 7, 15, 31, 63. Third differences: 8, 16, 32 (doubles).

Next: 64. Second difference = $63 + 64 = 127$. First difference = $121 + 127 = 248$. Next: $224 + 248 = 472$.

Answer: 472

15. Question: Find the next number: 5, 15, 35, 65, 105, 155, __

Solution:

Differences: 10, 20, 30, 40, 50. Differences increase by 10. Next difference = 60.

Next: $155 + 60 = 215$.

Answer: 215

16. Question: Find the next number: 2, 7, 17, 37, 77, 157, __

Solution:

Differences: 5, 10, 20, 40, 80 (doubles). Next difference = 160.

Next: $157 + 160 = 317$.

Answer: 317

17. Question: Find the next number: 1, 4, 12, 36, 108, 324, __

Solution:

Each number is multiplied by 3: $1 \times 3 = 4$, $4 \times 3 = 12$, $12 \times 3 = 36$, $36 \times 3 = 108$, $108 \times 3 = 324$.

Next: $324 \times 3 = 972$.

Answer: 972

18. Question: Find the next number: 3, 8, 22, 60, 164, 446, __

Solution:

Each number is multiplied by 3 and subtracted by a sequence: $3 \times 3 - 1 = 8$, $8 \times 3 - 2 = 22$, $22 \times 3 - 6 = 60$, $60 \times 3 - 16 = 164$, $164 \times 3 - 46 = 446$.

Subtractions: 1, 2, 6, 16, 46 (+1, $\times 3$). Next: 136. Next: $446 \times 3 - 136 = 1202$.

Answer: 1202

19. Question: Find the next number: 2, 5, 13, 35, 94, 248, __

Solution:

Each number follows: $2 \times 2 + 1 = 5$, $5 \times 3 - 2 = 13$, $13 \times 3 - 4 = 35$, $35 \times 3 - 11 = 94$, $94 \times 3 - 34 = 248$.

Subtractions: 0, 2, 4, 11, 34 (Fibonacci-like). Next: 88. Next: $248 \times 3 - 88 = 656$.

Answer: 656

20. Question: Find the next number: 100, 98, 92, 80, 62, 38, __

Solution:

Differences: -2, -6, -12, -18, -24. Second differences: -4, -6, -6, -6. Next difference = -30.

Next: $38 - 30 = 8$.

Answer: 8

Shape Pattern Questions (21?40)

21. Question: A square rotates 90° clockwise, doubles small squares inside, and alternates colors (Red, Blue): 1 (Red), 2 (Blue), 4 (Red), __. What is the next number, color, and rotation?

Solution:

Squares: 1, 2, 4, 8 (doubles). Colors: Red, Blue, Red, Blue. Rotation: $90^\circ \times 4 = 360^\circ = 0^\circ$.

Answer: 8 squares, Blue, 0° rotation

22. Question: A triangle rotates 120° anticlockwise, adds 1 side, and cycles fills (Filled, Empty, Half): Triangle (Filled), Quadrilateral (Empty), Pentagon (Half), __. What is the next shape, fill, and rotation?

Solution:

Sides: 3, 4, 5, 6 (Hexagon). Fill: Filled, Empty, Half, Filled. Rotation: $120^\circ \times 4 = 480^\circ = 120^\circ$.

Answer: Hexagon, Filled, 120° anticlockwise

23. Question: A circle doubles sectors and alternates fill and color (Red Filled, Blue Empty): 2 (Red Filled), 4 (Blue Empty), 8 (Red Filled), __. What is the next number, fill, and color?

Solution:

Sectors: 2, 4, 8, 16 (doubles). Fill/Color: Red Filled, Blue Empty, Red Filled, Blue Empty.

Answer: 16 sectors, Blue Empty

24. Question: A pentagon rotates 72° clockwise, increases sides by 2, and cycles colors (Red, Blue, Green): Pentagon (Red), Heptagon (Blue), Nonagon (Green), __. What is the next shape, color, and rotation?

Solution:

Sides: 5, 7, 9, 11 (11-gon). Colors: Red, Blue, Green, Red. Rotation: $72^\circ \times 4 = 288^\circ$.

Answer: 11-gon, Red, 288° clockwise

25. Question: A sequence alternates shapes (Circle, Square), doubles count, and cycles colors (Red, Blue): 1 Circle (Red), 2 Squares (Blue), 4 Circles (Red), ___. What is the next shape, number, and color?

Solution:

Shapes: Circle, Square, Circle, Square. Number: 1, 2, 4, 8. Colors: Red, Blue, Red, Blue.

Answer: 8 Squares, Blue

26. Question: A rectangle triples length, halves width, and rotates 90° anticlockwise: 3×4 , 9×2 , 27×1 , ___. What is the next rectangle and rotation?

Solution:

Length: 3, 9, 27, 81. Width: 4, 2, 1, 0.5. Rotation: $90^\circ \times 4 = 360^\circ = 0^\circ$.

Answer: 81×0.5 , 0° rotation

27. Question: A sequence of triangles adds 2, cycles fills (Filled, Empty), and rotates 60° clockwise: 2 (Filled), 4 (Empty), 6 (Filled), ___. What is the next number, fill, and rotation?

Solution:

Number: 2, 4, 6, 8. Fill: Filled, Empty, Filled, Empty. Rotation: $60^\circ \times 4 = 240^\circ$.

Answer: 8 triangles, Empty, 240° clockwise

28. Question: A square increases side by 3, adds 1 small square, and cycles colors (Red, Blue, Green): 3×3 (1 Red), 6×6 (2 Blue), 9×9 (3 Green), ___. What is the next square, number, and color?

Solution:

Side: 3, 6, 9, 12. Squares: 1, 2, 3, 4. Colors: Red, Blue, Green, Red.

Answer: 12×12 , 4 squares, Red

29. Question: A sequence cycles shapes (Circle, Square, Triangle), doubles count, and alternates fills: 1 Circle (Filled), 2 Squares (Empty), 4 Triangles (Filled), ___. What is the next shape, number, and fill?

Solution:

Shapes: Circle, Square, Triangle, Circle. Number: 1, 2, 4, 8. Fill: Filled, Empty, Filled, Empty.

Answer: 8 Circles, Empty

30. Question: A circle quadruples sectors, rotates 45° clockwise, and cycles colors (Red, Blue): 4 (Red), 16 (Blue), 64 (Red), ___. What is the next number, color, and rotation?

Solution:

Sectors: 4, 16, 64, 256 ($\times 4$). Colors: Red, Blue, Red, Blue. Rotation: $45^\circ \times 4 = 180^\circ$.

Answer: 256 sectors, Blue, 180° clockwise

31. Question: A sequence alternates shapes (Star, Circle), triples count, and cycles fills (Filled, Empty): 1 Star (Filled), 3 Circles (Empty), 9 Stars (Filled), ___. What is the next shape, number, and fill?

Solution:

Shapes: Star, Circle, Star, Circle. Number: 1, 3, 9, 27. Fill: Filled, Empty, Filled, Empty.

Answer: 27 Circles, Empty

32. Question: A hexagon rotates 60° anticlockwise, adds 3 sides, and cycles colors (Red, Blue, Green): Hexagon (Red), Nonagon (Blue), 12-gon (Green), ___. What is the next shape, color, and rotation?

Solution:

Sides: 6, 9, 12, 15 (15-gon). Colors: Red, Blue, Green, Red. Rotation: $60^\circ \times 4 = 240^\circ$.

Answer: 15-gon, Red, 240° anticlockwise

33. Question: A sequence of squares triples small squares and alternates colors (Red, Blue): 3 (Red), 9 (Blue), 27 (Red), ___. What is the next number, color, and rotation (90° clockwise)?

Solution:

Squares: 3, 9, 27, 81 ($\times 3$). Colors: Red, Blue, Red, Blue. Rotation: $90^\circ \times 4 = 360^\circ = 0^\circ$.

Answer: 81 squares, Blue, 0° rotation

34. Question: A triangle doubles sides, rotates 120° clockwise, and cycles fills (Filled, Empty, Half): 3 (Filled), 6 (Empty), 12 (Half), ___. What is the next shape, fill, and rotation?

Solution:

Sides: 3, 6, 12, 24. Fill: Filled, Empty, Half, Filled. Rotation: $120^\circ \times 4 = 480^\circ = 120^\circ$.

Answer: 24-gon, Filled, 120° clockwise

35. Question: A sequence cycles shapes (Circle, Square, Triangle), quadruples count, and cycles colors (Red, Blue, Green): 1 Circle (Red), 4 Squares (Blue), 16 Triangles (Green), ___. What is the next shape, number, and color?

Solution:

Shapes: Circle, Square, Triangle, Circle. Number: 1, 4, 16, 64. Colors: Red, Blue, Green, Red.

Answer: 64 Circles, Red

36. Question: A rectangle doubles length, adds 2 to width, and rotates 90° clockwise: 2×3 , 4×5 , 8×7 , ___. What is the next rectangle and rotation?

Solution:

Length: 2, 4, 8, 16. Width: 3, 5, 7, 9. Rotation: $90^\circ \times 4 = 360^\circ = 0^\circ$.

Answer: 16×9 , 0° rotation

37. Question: A sequence of circles adds 3, cycles fills (Filled, Empty, Half), and rotates 45° anticlockwise: 3 (Filled), 6 (Empty), 9 (Half), ___. What is the next number, fill, and rotation?

Solution:

Number: 3, 6, 9, 12. Fill: Filled, Empty, Half, Filled. Rotation: $45^\circ \times 4 = 180^\circ$.

Answer: 12 circles, Filled, 180° anticlockwise

38. Question: A pentagon adds 4 sides, rotates 36° clockwise, and alternates colors (Red, Blue): 5 (Red), 9 (Blue), 13 (Red), ___. What is the next shape, color, and rotation?

Solution:

Sides: 5, 9, 13, 17 (17-gon). Colors: Red, Blue, Red, Blue. Rotation: $36^\circ \times 4 = 144^\circ$.

Answer: 17-gon, Blue, 144° clockwise

39. Question: A sequence alternates shapes (Triangle, Square), triples count, and cycles colors (Red, Blue): 1 Triangle (Red), 3 Squares (Blue), 9 Triangles (Red), __. What is the next shape, number, and color?

Solution:

Shapes: Triangle, Square, Triangle, Square. Number: 1, 3, 9, 27. Colors: Red, Blue, Red, Blue.

Answer: 27 Squares, Blue

40. Question: A circle increases radius by 3, rotates 60° anticlockwise, and cycles fills (Filled, Empty): r=3 (Filled), r=6 (Empty), r=9 (Filled), __. What is the next radius, fill, and rotation?

Solution:

Radius: 3, 6, 9, 12. Fill: Filled, Empty, Filled, Empty. Rotation: $60^\circ \times 4 = 240^\circ$.

Answer: r=12, Empty, 240° anticlockwise

Analogy Questions (41?60)

41. Question: 2 : 12 :: 5 : __

Solution:

$2 \times (2 + 4) = 12$. Similarly, $5 \times (5 + 4) = 45$.

Answer: 45

42. Question: Circle (Red Filled) : Square (Blue Empty) :: Triangle (Green Half) : __

Solution:

Circle to Square (shape), Red to Blue (color), Filled to Empty (state). Triangle to Square, Green to Red, Half to Filled.

Answer: Square (Red Filled)

43. Question: 3 : 24 :: 4 : __

Solution:

$3 \times (3 + 5) = 24$. Similarly, $4 \times (4 + 5) = 36$.

Answer: 36

44. Question: Filled Circle (3) : Empty Square (4) :: Half Triangle (3) : __

Solution:

Filled to Empty (state), Circle to Square (shape +1 side), 3 to 4 (sides). Half to Filled, Triangle to Square, 3 to 4.

Answer: Filled Square (4)

45. Question: 5 : 60 :: 6 : __

Solution:

$5 \times (5 + 7) = 60$. Similarly, $6 \times (6 + 7) = 78$.

Answer: 78

46. Question: Star (5 Red) : Circle (0 Blue) :: Pentagon (5 Green) : ____

Solution:

Star to Circle (shape, sides 5?0), Red to Blue (color). Pentagon to Circle, Green to Red.

Answer: Circle (0 Red)

47. Question: 2 : 20 :: 3 : ____

Solution:

$2 \times (2 + 8) = 20$. Similarly, $3 \times (3 + 8) = 33$.

Answer: 33

48. Question: Red Filled : Blue Empty :: Green Half : ____

Solution:

Red to Blue (color), Filled to Empty (state). Green to Red, Half to Filled.

Answer: Red Filled

49. Question: 4 : 40 :: 5 : ____

Solution:

$4 \times (4 + 6) = 40$. Similarly, $5 \times (5 + 6) = 55$.

Answer: 55

50. Question: Triangle (3 Filled) : Square (4 Empty) :: Pentagon (5 Half) : ____

Solution:

Triangle to Square (+1 side), Filled to Empty. Pentagon to Hexagon, Half to Filled.

Answer: Hexagon (6 Filled)

51. Question: 1 : 6 :: 3 : ____

Solution:

$1 \times (1 + 5) = 6$. Similarly, $3 \times (3 + 5) = 24$.

Answer: 24

52. Question: Filled Circle : Empty Square :: Half Triangle : ____

Solution:

Filled to Empty (state), Circle to Square (shape). Half to Filled, Triangle to Square.

Answer: Filled Square

53. Question: 6 : 84 :: 7 : ____

Solution:

$6 \times (6 + 8) = 84$. Similarly, $7 \times (7 + 8) = 105$.

Answer: 105

54. Question: Circle (0 Red) : Triangle (3 Blue) :: Square (4 Green) : ____

Solution:

Circle to Triangle (+3 sides), Red to Blue. Square to Heptagon (+3), Green to Red.

Answer: Heptagon (7 Red)

55. Question: 2 : 18 :: 4 : ____

Solution:

$$2 \times (2 + 7) = 18. \text{ Similarly, } 4 \times (4 + 7) = 44.$$

Answer: 44

56. Question: Red Circle (Filled) : Blue Square (Empty) :: Green Triangle (Half) : ____

Solution:

Red to Blue, Circle to Square, Filled to Empty. Green to Red, Triangle to Square, Half to Filled.

Answer: Red Square (Filled)

57. Question: 3 : 30 :: 5 : ____

Solution:

$$3 \times (3 + 7) = 30. \text{ Similarly, } 5 \times (5 + 7) = 60.$$

Answer: 60

58. Question: Square (4 Filled) : Circle (0 Empty) :: Triangle (3 Half) : ____

Solution:

Square to Circle (-4 sides), Filled to Empty. Triangle to Circle (-3), Half to Filled.

Answer: Circle (0 Filled)

59. Question: 4 : 48 :: 6 : ____

Solution:

$$4 \times (4 + 8) = 48. \text{ Similarly, } 6 \times (6 + 8) = 84.$$

Answer: 84

60. Question: Pentagon (5 Red) : Hexagon (6 Blue) :: Heptagon (7 Green) : ____

Solution:

Pentagon to Hexagon (+1 side), Red to Blue. Heptagon to Octagon, Green to Red.

Answer: Octagon (8 Red)

Matrix Pattern Questions (61-80)

61. Question: Matrix: [1, 6, 16; 2, 8, 20; 3, 10, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 6, 8, 10 (+2); 16, 20, x (+4). Next: $20 + 4 = 24$.

Answer: 24

62. Question: Matrix: [2, 12, 36; 3, 18, 54; 4, 24, ____]. What is the missing number?

Solution:

Each row: $2 \times 1, 2 \times 6, 2 \times 18$; $3 \times 1, 3 \times 6, 3 \times 18$; $4 \times 1, 4 \times 6, 4 \times 18$.

Next: $4 \times 18 = 72$.

Answer: 72

63. Question: Matrix: [Red Filled, Blue Empty, Green Half; Blue Empty, Green Half, Red Filled;

Green Half, Red Filled, ____]. What is the missing state?

Solution:

Each column shifts down: Red Filled ? Blue Empty ? Green Half; Blue Empty ? Green Half ? Red Filled; Green Half ? Red Filled ? Blue Empty.

Answer: Blue Empty

64. Question: Matrix: [1, 5, 14; 2, 8, 20; 3, 11, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 5, 8, 11 (+3); 14, 20, x (+6). Next: $20 + 6 = 26$.

Answer: 26

65. Question: Matrix: [Circle Red, Square Blue, Triangle Green; Square Blue, Triangle Green, Circle Red; Triangle Green, Circle Red, ____]. What is the missing shape?

Solution:

Each column shifts down: Circle Red ? Square Blue ? Triangle Green; Square Blue ? Triangle Green ? Circle Red; Triangle Green ? Circle Red ? Square Blue.

Answer: Square Blue

66. Question: Matrix: [4, 16, 40; 5, 20, 50; 6, 24, ____]. What is the missing number?

Solution:

Each row: 4×1 , 4×4 , 4×10 ; 5×1 , 5×4 , 5×10 ; 6×1 , 6×4 , 6×10 .

Next: $6 \times 10 = 60$.

Answer: 60

67. Question: Matrix: [1, 7, 19; 2, 10, 25; 3, 13, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 7, 10, 13 (+3); 19, 25, x (+6). Next: $25 + 6 = 31$.

Answer: 31

68. Question: Matrix: [Filled Circle, Empty Square, Half Triangle; Empty Square, Half Triangle, Filled Circle; Half Triangle, Filled Circle, ____]. What is the missing shape?

Solution:

Each column shifts down: Circle ? Square ? Triangle; Square ? Triangle ? Circle; Triangle ? Circle ? Square.

Answer: Empty Square

69. Question: Matrix: [3, 12, 30; 4, 16, 40; 5, 20, ____]. What is the missing number?

Solution:

Each row: 3×1 , 3×4 , 3×10 ; 4×1 , 4×4 , 4×10 ; 5×1 , 5×4 , 5×10 .

Next: $5 \times 10 = 50$.

Answer: 50

70. Question: Matrix: [1, 4, 16; 2, 8, 32; 3, 12, ____]. What is the missing number?

Solution:

Each row: 1×4^1 , 1×4^2 ; 2×4^1 , 2×4^2 ; 3×4^1 , 3×4^2 .

Next: $3 \times 4^2 = 48$.

Answer: 48

71. Question: Matrix: [Star Red, Circle Blue, Square Green; Circle Blue, Square Green, Star Red; Square Green, Star Red, ____]. What is the missing shape?

Solution:

Each column shifts down: Star Red ? Circle Blue ? Square Green; Circle Blue ? Square Green ? Star Red; Square Green ? Star Red ? Circle Blue.

Answer: Circle Blue

72. Question: Matrix: [2, 10, 28; 3, 15, 42; 4, 20, ____]. What is the missing number?

Solution:

Each row: $2 \times 1, 2 \times 5, 2 \times 14; 3 \times 1, 3 \times 5, 3 \times 14; 4 \times 1, 4 \times 5, 4 \times 14$.

Next: $4 \times 14 = 56$.

Answer: 56

73. Question: Matrix: [1, 6, 18; 2, 8, 22; 3, 10, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 6, 8, 10 (+2); 18, 22, x (+4). Next: $22 + 4 = 26$.

Answer: 26

74. Question: Matrix: [Red Filled, Green Empty, Blue Half; Green Empty, Blue Half, Red Filled; Blue Half, Red Filled, ____]. What is the missing state?

Solution:

Each column shifts down: Red Filled ? Green Empty ? Blue Half; Green Empty ? Blue Half ? Red Filled; Blue Half ? Red Filled ? Green Empty.

Answer: Green Empty

75. Question: Matrix: [4, 20, 56; 5, 25, 70; 6, 30, ____]. What is the missing number?

Solution:

Each row: $4 \times 1, 4 \times 5, 4 \times 14; 5 \times 1, 5 \times 5, 5 \times 14; 6 \times 1, 6 \times 5, 6 \times 14$.

Next: $6 \times 14 = 84$.

Answer: 84

76. Question: Matrix: [1, 5, 17; 2, 8, 23; 3, 11, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 5, 8, 11 (+3); 17, 23, x (+6). Next: $23 + 6 = 29$.

Answer: 29

77. Question: Matrix: [2, 12, 36; 3, 18, 54; 4, 24, ____]. What is the missing number?

Solution:

Each row: $2 \times 1, 2 \times 6, 2 \times 18; 3 \times 1, 3 \times 6, 3 \times 18; 4 \times 1, 4 \times 6, 4 \times 18$.

Next: $4 \times 18 = 72$.

Answer: 72

78. Question: Matrix: [Filled Star, Empty Circle, Half Square; Empty Circle, Half Square, Filled Star; Half Square, Filled Star, ____]. What is the missing shape?

Solution:

Each column shifts down: Star ? Circle ? Square; Circle ? Square ? Star; Square ? Star ? Circle.

Answer: Empty Circle

79. Question: Matrix: [3, 15, 45; 4, 20, 60; 5, 25, ____]. What is the missing number?

Solution:

Each row: 3×1 , 3×5 , 3×15 ; 4×1 , 4×5 , 4×15 ; 5×1 , 5×5 , 5×15 .

Next: $5 \times 15 = 75$.

Answer: 75

80. Question: Matrix: [1, 7, 22; 2, 10, 28; 3, 13, ____]. What is the missing number?

Solution:

Columns: 1, 2, 3 (+1); 7, 10, 13 (+3); 22, 28, x (+6). Next: $28 + 6 = 34$.

Answer: 34

Odd One Out Questions (81?100)

81. Question: Which is the odd one out: 3, 12, 30, 66, 132, 242?

Solution:

Differences: 9, 18, 36, 66, 110. Second differences: 9, 18, 30, 44. 242 breaks the pattern.

Answer: 242

82. Question: Which is the odd one out: Circle (Red Filled), Square (Blue Empty), Triangle (Green Half), Pentagon (Red Filled), Star (Blue Half)?

Solution:

Circle, Square, Triangle, Pentagon are polygons. Star is not.

Answer: Star

83. Question: Which is the odd one out: 2, 7, 17, 37, 77, 152?

Solution:

Differences: 5, 10, 20, 40, 75. 152 breaks the doubling pattern.

Answer: 152

84. Question: Which is the odd one out: Filled Circle (Red), Empty Square (Blue), Half Triangle (Green), Filled Pentagon (Yellow), Empty Star (Red)?

Solution:

Star is not a polygon; others are.

Answer: Empty Star

85. Question: Which is the odd one out: 4, 12, 30, 66, 132, 264?

Solution:

Differences: 8, 18, 36, 66, 132. Second differences: 10, 18, 30, 66. 264 fits. Recompute: 264 is correct.

Answer: None (all fit, but 4 is smallest)

86. Question: Which is the odd one out: Star (Filled Red), Circle (Empty Blue), Square (Half Green), Triangle (Filled Yellow), Pentagon (Empty Red)?

Solution:

Star is not a polygon; others are.

Answer: Star

87. Question: Which is the odd one out: 1, 5, 16, 44, 104, 224, 464?

Solution:

Differences: 4, 11, 28, 60, 120, 240. Second differences deviate at 464.

Answer: 464

88. Question: Which is the odd one out: Red Filled, Blue Empty, Green Half, Yellow Filled, Circle Empty?

Solution:

Circle Empty is a shape; others are color/state.

Answer: Circle Empty

89. Question: Which is the odd one out: 2, 5, 13, 35, 94, 248, 650?

Solution:

Each follows: $\times 3 - 1$, $\times 3 - 2$, etc. $650 \neq 248 \times 3 - 94$.

Answer: 650

90. Question: Which is the odd one out: Triangle (3 Red), Square (4 Blue), Pentagon (5 Green), Circle (0 Yellow), Hexagon (6 Red)?

Solution:

Circle has 0 sides; others are polygons.

Answer: Circle

91. Question: Which is the odd one out: 3, 9, 22, 46, 88, 158, 272?

Solution:

All fit pattern. No odd one out.

Answer: None

92. Question: Which is the odd one out: Blue Circle (Filled), Red Square (Empty), Green Triangle (Half), Yellow Pentagon (Filled), Star (Red Empty)?

Solution:

Star is not a polygon.

Answer: Star

93. Question: Which is the odd one out: 5, 15, 35, 65, 105, 155, 205?

Solution:

Differences: 10, 20, 30, 40, 50, 50. 205 breaks +10 pattern.

Answer: 205

94. Question: Which is the odd one out: Pentagon (Red Filled), Hexagon (Blue Empty), Heptagon (Green Half), Octagon (Yellow Filled), Circle (Red Empty)?

Solution:

Circle is not a polygon.

Answer: Circle

95. Question: Which is the odd one out: 2, 4, 11, 29, 74, 182, 446?

Solution:

$446 \neq 182 \times 3 - 104$.

Answer: 446

96. Question: Which is the odd one out: Filled Star (Red), Empty Circle (Blue), Half Square (Green), Filled Triangle (Yellow), Empty Pentagon (Green)?

Solution:

Star is not a polygon.

Answer: Star

97. Question: Which is the odd one out: 1, 4, 12, 36, 108, 324, 970?

Solution:

$970 \neq 324 \times 3$.

Answer: 970

98. Question: Which is the odd one out: Red Filled Circle, Blue Empty Square, Green Half Triangle, Yellow Filled Pentagon, Green Empty Star?

Solution:

Star is not a polygon.

Answer: Star

99. Question: Which is the odd one out: 3, 8, 22, 60, 164, 446, 1204?

Solution:

$1204 \neq 446 \times 3 - 136$.

Answer: 1204

100. Question: Which is the odd one out: Square (Red Filled), Triangle (Blue Empty), Pentagon (Green Half), Hexagon (Yellow Filled), Star (Red Half)?

Solution:

Star is not a polygon.

Answer: Star